Progressive exercises

E1 · Practice. Define on $\mathbb{R}$ the functions $f(x) = (2x-1)^5$ and $g(x) = e^{-x^2}$. Calculate $f'(x)$ and $g'(x)$.

E2 · Practice. Let f be the function defined on $\mathbb{R}$ by $f(x) = x^4 - 6x^2$. Determine the intervals on which f is convex or concave.

E3 · Practice. Let $\mathcal{C}$ be the graph of the exponential function. Determine an equation of its tangent at the point with abscissa 1. Using convexity, deduce an inequality valid for every real number x.

E4 · Practice. Prove that the equation $x^3 + 2x = 5$ has a unique real solution. Bracket this solution between two consecutive integers.

E5 · Practice. A bisection algorithm starts from the interval $[0,2]$ and halves its length at each step. Determine the minimum number of steps guaranteeing an interval of length at most 10^{-3}.

Textbook or past-paper work

Barbazo, p. 132, exercise 29. In class after the study of convexity: questions 1 and 2. At home: finish questions 3 to 5. The aim is to distinguish the sign of f', which governs the variations, from that of f'', which governs convexity.